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**COMMENTS OF THE VEHICLE GRID INTEGRATION COUNCIL ON DOCKET NO.
D.P.U. 25-189 & DOCKET NO. D.P.U. 25-188 PERTAINING TO THE DISTRIBUTION
UTILITY PETITIONS FOR PHASE IV (NATIONAL GRID) & III (EVERSOURCE) EV
PROGRAM PLAN PROPOSALS**

I. INTRODUCTION

The Vehicle-Grid Integration Council (VGIC) is a 501(c)(6) membership-based advocacy organization committed to advancing the role of vehicle-grid integration (VGI) through policy development, education, and research. VGIC supports the transition to decarbonized transportation and electric sectors by ensuring that the value of flexible EV charging and discharging is recognized, compensated, and aligned with a more reliable, affordable, and efficient electric grid.

Transportation electrification is a core element of Massachusetts' climate strategy, but its long-term success depends on keeping costs affordable for both EV drivers and all ratepayers. Electric vehicles reduce fuel and maintenance costs for customers while cutting emissions from the state's largest source of pollution. Furthermore, EV charging load increases grid asset utilization,

putting downward pressure on rates. Continued investment in charging infrastructure, particularly in underserved communities and for medium- and heavy-duty fleets, supports this transition while ensuring that the benefits of electrification are broadly shared.

Managed charging and bidirectional charging initiatives are critical to protecting affordability. By shifting EV charging to off-peak hours and aligning charging and discharging behaviors based on grid conditions, investments in managed charging programs and bidirectional charging technologies reduce the need for costly system upgrades, lowering costs for all customers, not just EV drivers. Importantly, VGI technology is evolving rapidly. From 2023-2026, utilities and customers have gained access to a wider range of interoperable chargers, vehicle capabilities, and software platforms, enabling more precise and cost-effective optimization of managed and bidirectional charging. This trend is anticipated to continue and even accelerate over the 2027–2030 program period.

VGIC appreciates the opportunity to submit comments on National Grid’s Phase IV EV Plan program and Eversource’s Phase III EV Plan program proposals for 2027–2030. Furthermore, VGIC looks forward to continued collaboration to ensure that EV and the related infrastructure investments advance affordability, equity, and system efficiency in the Commonwealth of Massachusetts.

II. SUMMARY

These comments present targeted recommendations to strengthen National Grid’s Phase IV and Eversource’s Phase III EV Plan program proposals for the 2027–2030 period. As EV adoption accelerates, the EV Plan programs should reflect the demonstrated system value of managed and bidirectional charging by aligning incentives, enrollment targets, and program design with current grid needs and avoided cost conditions. The recommendations below are intended to improve cost-effectiveness, ensure consistency across utilities, and fully leverage EVs, including those already in Massachusetts’ driveways and parking garages, as flexible grid resources.

Key Recommendations

- Direct both utilities to materially increase managed charging enrollment targets to better reflect projected EV adoption and the peak reduction potential identified in DOER-funded studies.
- Specific to National Grid’s proposal:
 - Approve National Grid’s proposal to make residential bidirectional chargers eligible for enhanced incentives to support early V2X market development and capture incremental peak reduction value.
 - Approve National Grid’s proposed \$10,000 incentive for fleet bidirectional charging applications and remove the proposed \$1 million funding cap, particularly for proven school bus V2G use cases.

- Approve National Grid’s managed charging programs with modifications that allow customers to access both system-level and distribution-level value streams where incremental benefits can be demonstrated.
- Require National Grid to align managed charging incentive levels with the most recent avoided cost projections.
- Specific to Eversource’s proposal:
 - Require Eversource to offer residential and fleet bidirectional charging incentives comparable to National Grid to ensure consistent market signals across the Commonwealth and unlock additional net benefit for ratepayers. Eversource should also encourage these customers to participate in Connected Solutions or, if that is deemed unsuitable, another performance-based program offering.
 - Require Eversource to design managed charging programs that address both ISO-NE system peak conditions and localized distribution constraints, with appropriate value stacking.
 - Require Eversource to align managed charging incentive levels with the most current avoided cost assumptions.

III. COMMENTS

The following comments address specific elements of the National Grid and Eversource proposals that would benefit from targeted modifications to enhance cost-effectiveness and long-term system value.

1) The Department Should Require National Grid and Eversource to Expand Their Ambition for Managed Charging Program Enrollments

Massachusetts has a goal to have roughly 900,000 electric vehicles on the road by 2030,¹ yet the managed charging program enrollment targets proposed by National Grid and Eversource fall well short of what is needed to capture the full system value of EVs as grid resources. National Grid’s proposal to enroll approximately 64,000 EVs and Eversource’s proposal to enroll 11,500 EVs together represent about 8 percent of the projected EV fleet in 2030. This level of ambition may be sufficient for early pilots, but it is not commensurate with the scale of the opportunity, the timeline of the proposed EV Plan programs, or the grid challenges the Commonwealth is seeking to address.

This under-ambition stands in contrast to the findings of the DOER-funded Technical Potential of Load Management study conducted by Energy and Environmental Economics (E3) for the Massachusetts Department of Energy Resources. As noted above, that analysis identifies EV load flexibility as the largest and most cost-effective demand-side resource available to Massachusetts over the next several decades.

¹ Massachusetts Executive Office Energy and Environmental Affairs, June, 2022, Massachusetts Clean Energy and Climate Plan for 2025 and 2030, page 31.

Given these findings, the DPU should require both utilities to revisit and materially increase their managed charging enrollment targets. Doing so would better align program design with the Commonwealth's peak reduction and system planning objectives, send a clearer signal to the market about the long-term role of EVs as grid resources, and ensure that ratepayers capture the full value of investments already being made to electrify transportation.

2) Recommendations Specific to National Grid

a. The Department of Public Utilities (DPU) Should Approve National Grid's Proposal to Make Bidirectional Chargers for Residential Customers Eligible for the Enhanced Incentive

National Grid's proposal to extend enhanced incentives to residential bidirectional EV chargers is well aligned with Massachusetts' affordability, reliability, and grid modernization goals. These incentives help overcome early adopter barriers for customers and support market development and transition for bidirectional charging solutions. By qualifying bidirectional chargers for enhanced incentives, the DPU would send a clear signal to the market that V2X is not simply a customer amenity, but a grid resource that can materially reduce peak demand, defer infrastructure investments, and lower costs for all ratepayers.

While managed charging is a proven, low-cost strategy that shifts EV load away from peak hours and delivers immediate capacity benefits, multiple studies show that bidirectional charging behavior significantly amplifies these benefits. The Massachusetts DOER Peak Potential report finds that EV load management, including bidirectional charging, represents one of the highest-value, no-regrets flexibility resources available, contributing up to 6.5 GW of peak load reduction by 2050. Achieving this level of impact will require V2X equipment deployment, tapping into the nearly 150% greater peak reduction per vehicle from bidirectional charging compared to managed charging alone.² By reducing coincident peak demand more deeply, bidirectional charging directly lowers capacity, transmission, and distribution system costs that would otherwise be borne by all customers.

These findings are reinforced by detailed modeling from the Union of Concerned Scientists (UCS), which finds that bidirectional charging delivers materially greater peak and capacity value than managed charging alone. In its California analysis, UCS shows that shifting EV participation from one-way managed charging to bidirectional charging nearly doubles total system savings at moderate participation levels, and that scenarios with significant V2X deployment deliver more than 60 percent greater grid value than V1G-only scenarios.³ These incremental benefits are driven primarily by deeper peak load reductions enabled by dispatchable

² Massachusetts Department of Energy Resources, December 2025, *Peak Potential: Reducing energy costs and empowering consumers with load management and virtual power plants*, page 40.

³ Union of Concerned Scientists, June 2025, *Harnessing the Power of Electric Vehicles Vehicle-Grid Integration for a Cleaner, Cheaper, More Reliable California Electricity System*, page 15.

EV exports during the most constrained hours, allowing EVs to function as short-duration storage resources rather than passive load-shifting assets. Together, these results demonstrate that supporting early consumer adoption of V2X through enhanced incentives can unlock substantially greater, cost-effective grid value than managed charging alone.

Additionally, the DPU can be confident that bidirectional charging is transitioning rapidly from pilot programs to scalable commercial deployment. Recent announcements from both automakers and charger manufacturers indicate that bidirectional capability is moving into standard production offerings, rather than bespoke demonstrations and science projects. For example, Ford, GM (Chevrolet, Cadillac, and GMC vehicles), Nissan, and Kia all currently market grid-parallel and grid-supportive bidirectional charging features. Tesla recent launched its Powershare Grid Support program in Texas, representing the first vehicle-to-grid offering from a major automaker at scale to leverage lower-cost AC-based product architecture.⁴ Additional announcements from Toyota, Honda, Rivian, Lucid, BMW, VW, and Stellantis creates a near-universal suite of bidirectional charging offerings from light-duty automakers in the coming 2-3 years. On the charger side, dcbel, Wallbox, Ford, GM Energy, and Tesla offer bidirectional charging solutions while Emporia, ChargePoint, Autel, and SolarEdge have shared plans to launch bidirectional charging solutions, with Enphase Energy having recently detailed plans to begin volume production of its IQ Bidirectional EV Charging Platform in Q4 2026.⁵ Together, these developments signal that the 2027–2030 program period will coincide with significantly broader availability of bidirectional charging vehicles and chargers. Approving National Grid’s proposal now ensures that Massachusetts’ incentive framework keeps pace with this market evolution, avoids locking in one-way solutions, and positions the Commonwealth to capture the full peak-reduction and cost-containment benefits of EVs as grid assets.

b. The Department Should Approve National Grid’s Proposal to Provide \$10,000 Incentive for Fleet Bidirectional Charging Applications and Not Limit the Funding to \$1M

As discussed above, bidirectional charging delivers materially greater system value than managed charging alone by enabling vehicles to export power during peak conditions, reducing capacity needs and deferring costly grid investments. These benefits are especially pronounced in fleet applications, where vehicles have predictable duty cycles, long dwell times, and centralized charging infrastructure. Providing a \$10,000 incentive for fleet bidirectional charging applications reflects the higher upfront costs and integration complexity associated with bidirectional charging, while unlocking resources that deliver outsized, cost-effective peak reduction and reliability benefits for all ratepayers.

⁴ See TESLARATI, Tesla launches Cybertruck vehicle-to-grid program in Texas, available at <https://www.teslarati.com/tesla-launches-cybertruck-v2g-powershare-texas/>.

⁵ See Solar Power World, Enphase to bring bidirectional EV charger to market next year available at <https://www.solarpowerworldonline.com/2025/09/enphase-to-bring-bidirectional-ev-charger-to-market-next-year/>.

Within the medium- and heavy-duty (MHD) segment, school buses represent the most mature and proven V2G use case. School bus V2G deployments have been operating for several years across multiple states, supported by commercially available vehicles, bidirectional chargers, and aggregation platforms. Importantly, the market has made meaningful progress toward an interoperable ecosystem based on ISO 15118-20, demonstrating that OEM-provided bidirectional school buses can operate in bidirectional mode across multiple charger manufacturers.

V2G school bus projects show strong alignment between grid needs and vehicle availability, making school buses among the lowest-risk, highest-value V2G investments available today. Other MHD use cases are also being evaluated for bidirectional charging, including the refuse sector, seasonal tour buses, street sweeping vehicles, and other specialty vehicles with duty cycles that align with grid needs. **The Department should not limit funding for fleet bidirectional charging to an arbitrary \$1 million cap.** Instead, it should allow these investments to scale where they provide clear system value, ensuring that proven school bus and other MHD V2G applications are fully leveraged to reduce long-term system costs and support grid reliability.

c. The Department Should Approve the Managed Charging Programs Proposed by National Grid with Modifications

The managed charging programs proposed by National Grid represent a reasonable and cost-effective approach to integrating growing EV load in a manner that supports system reliability and affordability. As discussed above, managed charging is justified as a least-cost grid resource based on its ability to reduce coincident peak demand and avoid or defer capacity and distribution system investments. To sustain and scale these programs, the Department should affirm continued customer participation pathways that rely on both EVSE-based controls and vehicle telematics to measure performance, and support a mix of passive and active managed charging approaches. The Department should approve these programs subject to targeted modifications that better align incentives with demonstrated system value and evolving avoided-cost conditions.

Modification #1: Allow customers to access both system-level and distribution-level value streams where incentives reflect incremental benefits.

- The Department should recognize the innovative managed charging use cases in National Grid's proposal that target both system peak and distribution peak conditions, while clarifying that certain customers may receive compensation for both services when managed charging delivers incremental and clearly distinguishable benefits. Managed charging can provide separate value streams, for example, reducing regional system peak demand while also alleviating localized distribution constraints, particularly for secondary transformers as the near-term bottleneck, when events are differentiated by

timing, location, or operational purpose.⁶ Where National Grid can demonstrate that program design, measurement, and settlement appropriately distinguish between these services and avoid double-counting of the same load reduction, customers should be permitted to access both value streams. This approach aligns compensation with actual grid value, preserves cost-effectiveness, and strengthens incentives for participation without overpaying for the same load shift.

Modification #2: Update managed charging incentive levels to reflect the most recent avoided-cost projections.

- National Grid’s proposed incentive levels rely on avoided-cost assumptions that may not fully reflect current or forward-looking system conditions. Given rising capacity costs, increasing distribution constraints, and the growing importance of flexible load as a planning resource, the Department should require that managed charging incentives be updated to reflect the most recent avoided-cost projections. Aligning incentive levels with updated avoided costs will ensure that compensation remains commensurate with system value, strengthen the economic signal for participation, and better position managed charging programs to scale in line with the Commonwealth’s electrification and peak-reduction objectives.

3) Recommendations Specific to Eversource

a. Eversource Should Offer Similar Incentives to National Grid for Bidirectional Charging Residential and Fleet Applications

Eversource should offer incentives for residential and fleet bidirectional charging applications that are comparable to those proposed by National Grid. As discussed above, bidirectional charging delivers greater system value than managed charging alone by reducing peak demand, deferring infrastructure investments, and lowering long-term costs for all ratepayers. Aligning incentive levels and program design across utilities will help ensure consistent market signals, reduce customer and vendor confusion, and support efficient statewide deployment of V2X technologies. Program consistency across the Commonwealth is especially important for fleet operators, vehicle and charger manufacturers, and solution providers that operate across service territories, and will accelerate adoption of proven, cost-effective bidirectional charging resources where they provide clear grid value.

Moreover, while Eversource asserts in its filing that the existing Connected Solutions demand response program is not well suited to accommodate residential bidirectional charging and

⁶ For analysis of the value of managed charging to address distribution system constraints see Smart Electric Power Alliance, July 2025, Demonstrating of Utility Smart Charge Management for Multiple Benefit Streams and The Brattle Group, January 2026, Demonstrating the Full Value of Managed Electric Vehicle Charging Based on a Real-World Trial of EnergyHub’s EV Solution. These studies find annual savings potential of up to \$400 per vehicle for managed charging solutions that target distribution system value.

exports, this position stands in contrast to National Grid’s proposal, which requires residential customers receiving enhanced incentives for bidirectional chargers to enroll in Connected Solutions. Given that Eversource does not view Connected Solutions as appropriate for residential exports, the Department should direct Eversource to either actively open the program to bidirectional charging customers or launch another stopgap solution, such as a dedicated program that captures the demonstrated system value of residential EVs with V2X capabilities.

b. The Department Should Approve the Managed Charging Programs Proposed by Eversource with Modifications

The managed charging programs proposed by Eversource represent an important step toward integrating growing EV load in a manner that supports system reliability and customer affordability. The Department should approve these programs, subject to targeted modifications that better align program design with demonstrated grid value and evolving system needs.

Modification #1: Address both ISO-NE system peak and distribution system peak conditions.

- The Department should require Eversource to follow National Grid’s approach by designing managed charging programs that address both regional system peak conditions and localized distribution system constraints where they exist, with a particular focus on avoiding upgrades to secondary transformers that can quickly become overloaded from EV neighborhood clusters. Managed charging can deliver distinct and stackable value streams depending on the timing, location, and purpose of load reductions, and program design should reflect these differences. Where incremental benefits can be demonstrated and double-compensation avoided, customers should be able to access both system-level and distribution-level value streams. Explicitly incorporating both use cases will better align managed charging with actual grid needs and unlock greater overall system value.

Modification #2: Update managed charging incentive levels to reflect current avoided cost projections.

- Similar to Modification #2 for National Grid discussed above, Eversource should be required to update its proposed managed charging incentives using the most recent avoided cost projections. As capacity, transmission, and distribution system costs continue to evolve, incentive levels should remain commensurate with the system value delivered. Aligning incentive levels with updated avoided costs will strengthen program cost-effectiveness, improve participation signals, and ensure that managed charging scales appropriately as a core grid resource.

IV. CONCLUSION

VGIC appreciates the opportunity to submit these comments and welcomes continued collaboration with the utilities, the Department of Public Utilities, and other stakeholders to

realize the environmental and economic benefits for all Massachusetts ratepayers from efficiently leveraging electric vehicles in ways that enhance grid performance.

Respectfully submitted,

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